OOAD MINI PROJECT

PHASE – 1

Multi-Level Parking Lot System

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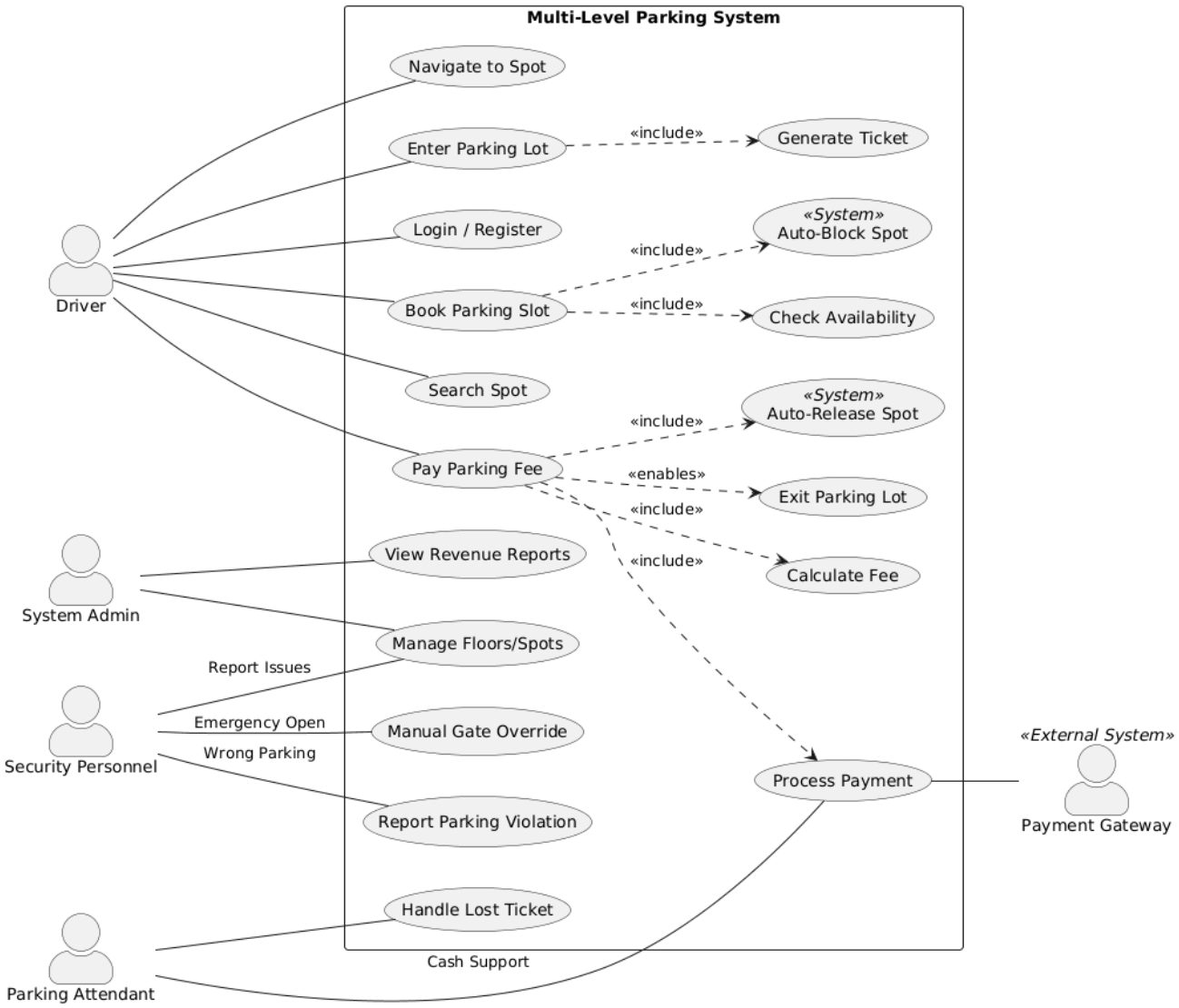
Use Case Diagram



Use Cases:

Primary Actor	Use Case ID	Use Case Name	System Automation / Logic	Description
Driver	UC-01	Login / Register	Validates user credentials.	Accesses the account to manage bookings and payments.
Driver	UC-02	Search Spot	Queries real-time floor data.	Searches for available spots by level or vehicle type.
Driver	UC-03	Filter by Floor	Optional query parameter.	[Extend] Narrows search to specific floors (1, 2, 3, etc.).
Driver	UC-04	Book Parking Slot	Auto-Block Spot	Triggers the system to change spot status to "Reserved".
System	UC-05	Check Availability	Internal validation check.	[Include] Ensures the spot is free before finalizing booking.
Driver	UC-06	Enter Parking Lot	Starts parking timer.	Driver arrives at entrance; triggers ticket generation.
System	UC-07	Generate Ticket	Creates digital record.	[Include] Maps vehicle ID to the entry time and spot.
Driver	UC-08	Navigate to Spot	Displays floor map/directions.	Guides the driver to the specific level and assigned stall.
Driver	UC-09	Pay Parking Fee	Auto-Release Spot	Successful payment triggers the spot reset to "Available".
System	UC-10	Calculate Fee	Mathematical logic.	[Include] Computes cost based on time and floor rates.
Payment Gateway	UC-11	Process Credit Card	External authorization.	Handles the digital transaction through the bank.
Attendant	UC-12	Process Cash	Updates payment state.	Attendant handles cash; system still triggers Auto-Release.

Primary Actor	Use Case ID	Use Case Name	System Automation / Logic	Description
Attendant	UC-13	Handle Lost Ticket	Manual data override.	Resolves issues when a driver has no entry record.
Admin	UC-14	Manage Infrastructure	Database configuration.	Adds/removes floors or spots from the active system.
Admin	UC-15	Manage Rate Cards	Pricing logic update.	Sets different costs for Premium vs. Economy floors.
Admin	UC-16	View Revenue Reports	Data aggregation.	Generates financial and occupancy statistics.
Driver	UC-17	Exit Parking Lot	End-of-session validation.	Departure is enabled once the spot status is "Available" (Released).